

EmbedUR Networks Training

Assessment – 4

Module – 7: Layer 3 – Ports and SNMP Protocol

&

Module – 8: Transport Layer Fundamentals

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1. Try Test-Connection and nslookup commands for below websites

www.google.com

www.facebook.com

www.amazon.com

www.github.com

www.cisco.com

```
kishore@kishore-virtual-machine:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.110
Name:   google.com
Address: 2404:6800:4007:815::200e

kishore@kishore-virtual-machine:~$ nslookup facebook.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   facebook.com
Address: 57.144.210.1
Name:   facebook.com
Address: 2a03:2880:f369:1:face:b00c:0:25de

kishore@kishore-virtual-machine:~$ nslookup github.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   github.com
Address: 20.207.73.82

kishore@kishore-virtual-machine:~$ nslookup cisco.com
Server:      127.0.0.53
Address:     127.0.0.53#53

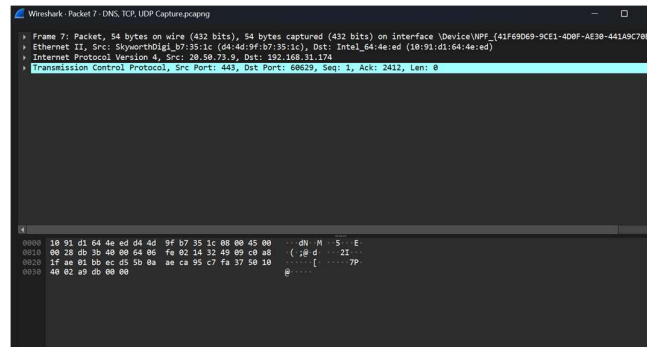
Non-authoritative answer:
Name:   cisco.com
Address: 72.163.4.185
Name:   cisco.com
Address: 2001:420:1101:1::185

kishore@kishore-virtual-machine:~$ nslookup amazon.com
Server:      127.0.0.53
Address:     127.0.0.53#53

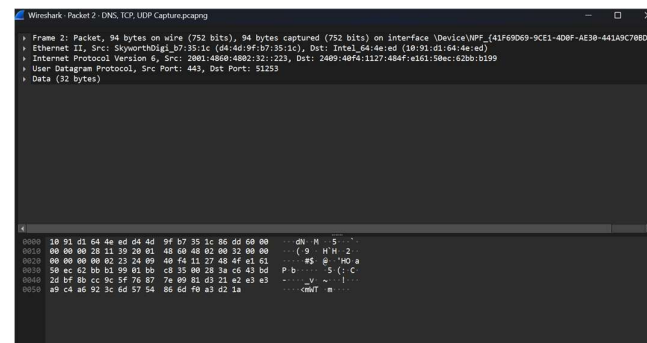
Non-authoritative answer:
Name:   amazon.com
Address: 98.82.161.185
Name:   amazon.com
Address: 98.87.170.74
```

2. Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header, packet flow, options and flags.

TCP:



UDP:



3. Explore traceroute/tracert for different websites eg:google.com and analyse the parameters in the output and explore different options for traceroute command

```
kishore@kishore-virtual-machine:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=71.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=58.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=48.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=60.8 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=112 time=51.5 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=112 time=46.2 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=112 time=61.5 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=112 time=48.0 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=112 time=52.8 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8014ms
rtt min/avg/max/mdev = 46.230/55.408/71.257/7.745 ms
kishore@kishore-virtual-machine:~$ traceroute ping 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 64 hops max
 1  192.168.31.1  3.304ms  1.752ms  2.500ms
 2  192.0.0.1  12.222ms  2.614ms  2.269ms
 3  192.0.0.1  39.739ms  38.156ms  39.038ms
 4  * * *
 5  192.0.0.1  48.052ms  63.330ms  57.817ms
 6  192.168.225.99  168.040ms  42.983ms  29.988ms
 7  192.168.83.28  39.185ms  27.136ms  42.020ms
 8  * * *
 9  * * *
10  209.85.175.48  49.382ms  56.669ms  42.797ms
11  * * *
12  8.8.8.8  70.266ms  39.472ms  42.364ms
kishore@kishore-virtual-machine:~$
```

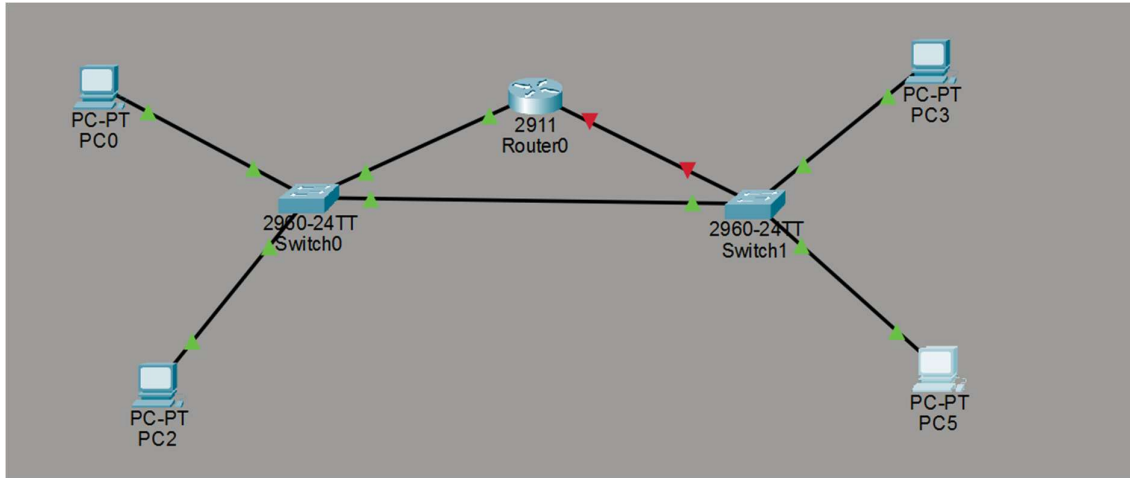
4. Use Cisco packet tracer for the below

Set up trunk ports between switches and try ping between different VLANs.

Change the native VLAN on a trunk port. Test for VLAN mismatches and troubleshoot.

Configure a management VLAN and assign an IP address for remote access. Test SSH or Telnet access to the switch.

Network Topology:



Switch - 1 Config:

Set VLAN Identifier Names:

```
Switch5
Physical Config CLI Attributes
Switch>
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
```


Set Port Types for every VLANs:



```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode trunk

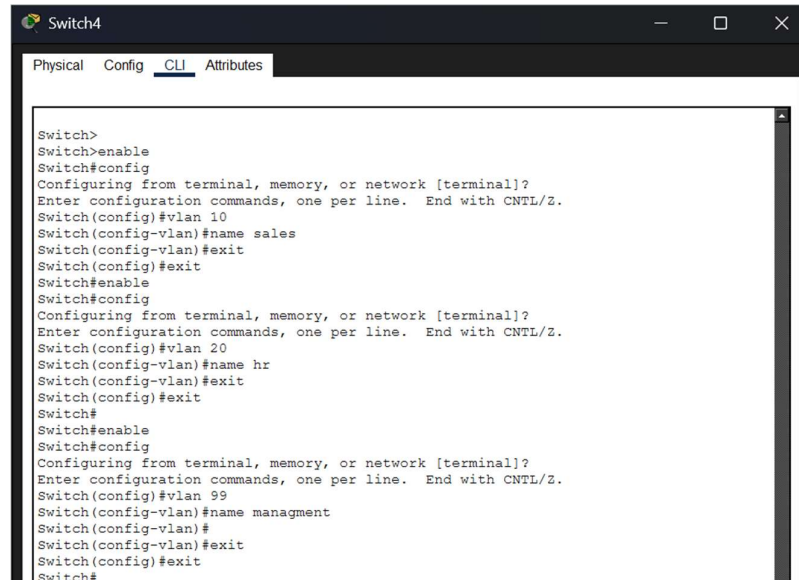
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
```

Copy Paste

Top

Switch -2 Config:

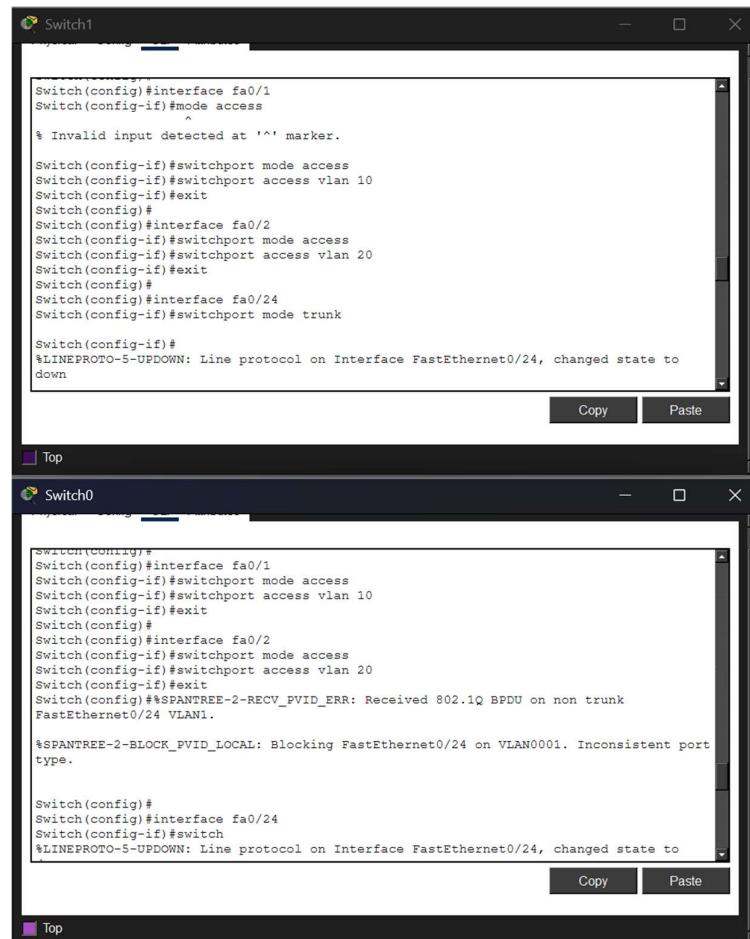
Create VLAN Identifiers:



```
Switch4
Physical Config CLI Attributes

Switch>
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#exit
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name management
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
```

Assign Access Ports:



```
Switch1
Switch(config)#interface fa0/1
Switch(config-if)#mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

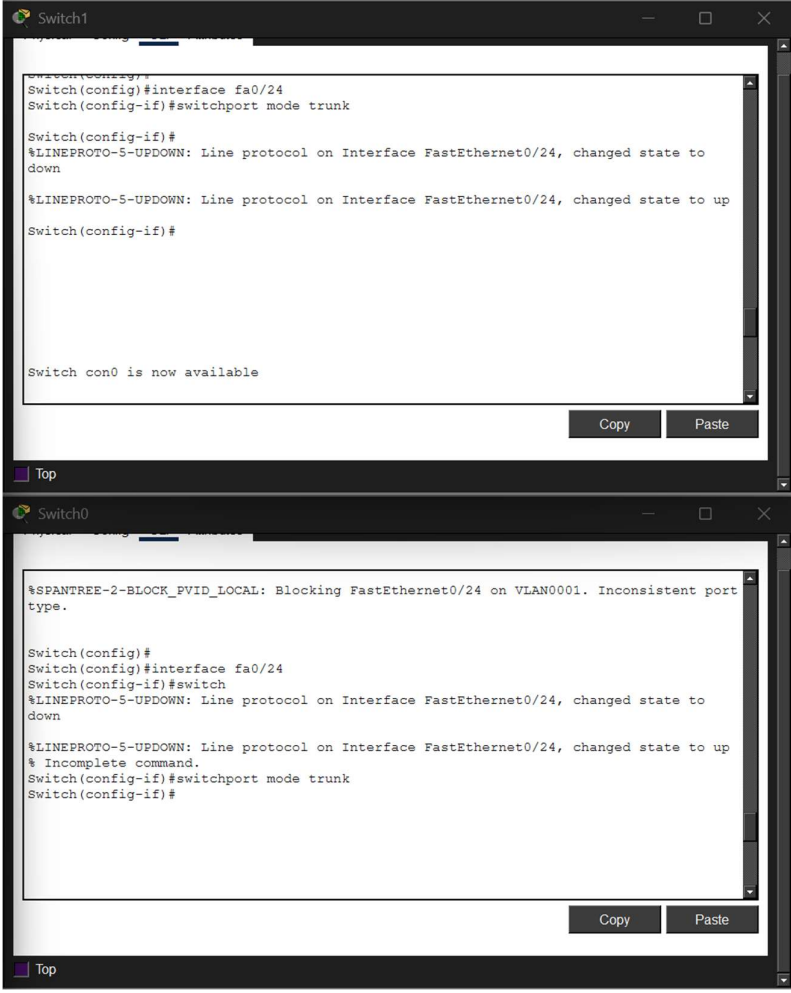
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
down
Copy Paste

Switch0
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#%SPANTREE-2-RECV_FVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/24 VLAN1.

%SPANTREE-2-BLOCK_FVID_LOCAL: Blocking FastEthernet0/24 on VLAN0001. Inconsistent port
type.

Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switch
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
down
Copy Paste
```

Assign Trunk Ports:



The image displays two screenshots of Cisco IOS command-line interfaces, one for Switch1 and one for Switch0, showing the configuration of a trunk port.

Switch1:

```
Switch1>enable
Switch1(config)#interface fa0/24
Switch1(config-if)#switchport mode trunk

Switch1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
Switch1(config-if)#

Switch con0 is now available
```

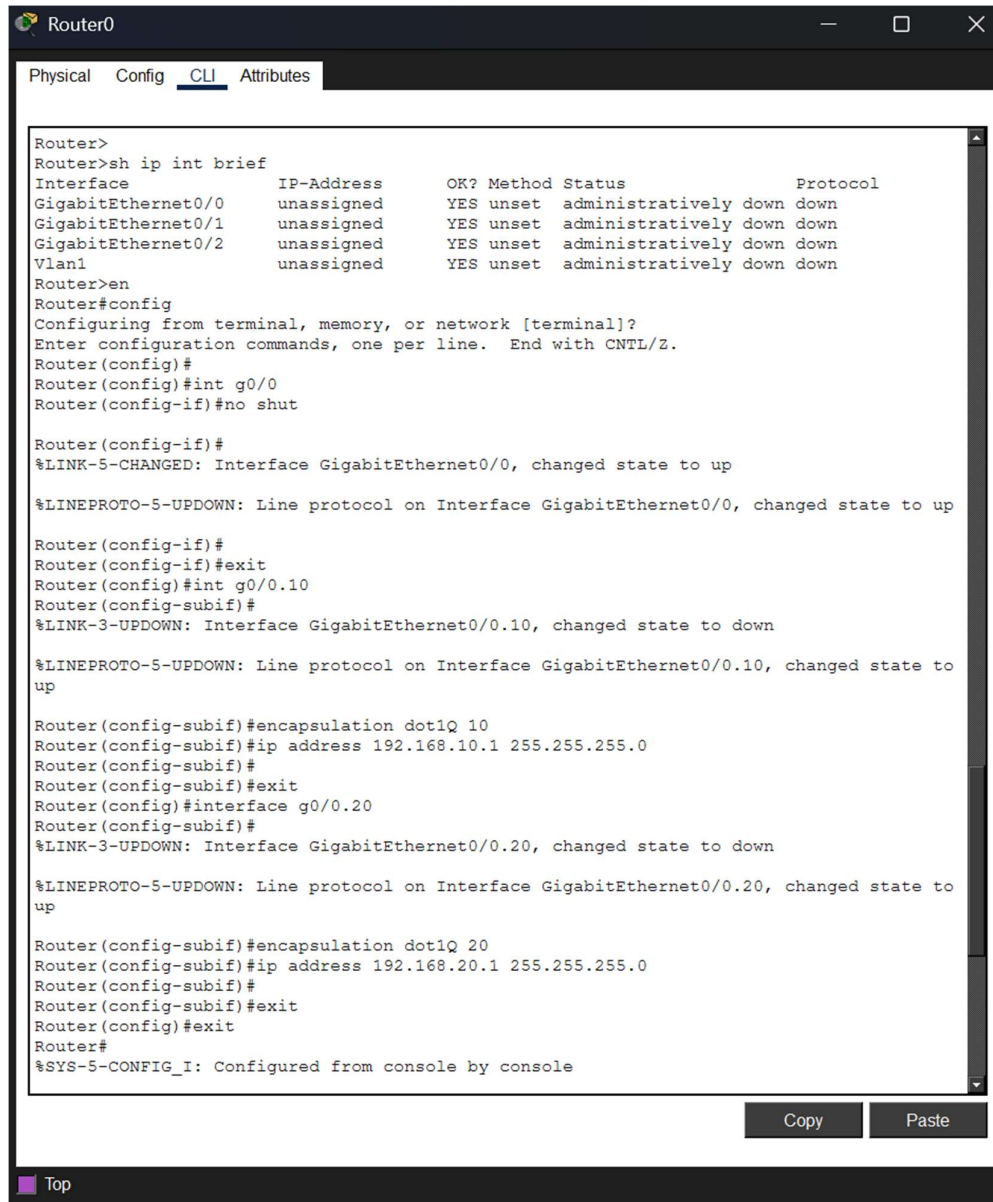
Switch0:

```
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/24 on VLAN0001. Inconsistent port
type.

Switch0(config)#
Switch0(config)#interface fa0/24
Switch0(config-if)#switch
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
% Incomplete command.
Switch0(config-if)#switchport mode trunk
Switch0(config-if)#
```

Router Config:



The screenshot shows a terminal window titled "Router0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of commands and their outputs. The commands include a brief interface status check, enabling configuration mode, configuring interface GigabitEthernet0/0, and configuring two subinterfaces (GigabitEthernet0/0.10 and GigabitEthernet0/0.20) with dot1Q encapsulation and IP addresses. The status output shows that the main interface and subinterfaces are administratively down but have link and line protocol status changes.

```
Router>
Router>sh ip int brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  administratively down  down
Vlan1                    unassigned      YES unset  administratively down  down
Router>en
Router>#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int g0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#int g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom of the terminal window, there are "Copy" and "Paste" buttons, and a "Top" link.

Configure Router to Switch Port as Trunk Port:

```
Switch>
Switch>en
Switch>#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping test:

The image displays four separate Command Prompt windows, each representing a different PC (PC0, PC1, PC2, PC3) in a network simulation. Each window shows the results of a ping test to a specific IP address.

- PC0:** Shows a ping test to 192.168.20.2. The results indicate that 4 packets were sent, 0 were received, and there was a 100% loss. The statistics show a minimum round trip time of 0ms, a maximum of 9ms, and an average of 4ms.
- PC1:** Shows a ping test to 192.168.20.3. The results indicate that 4 packets were sent, 4 were received, and there was a 0% loss. The statistics show a minimum round trip time of 1ms, a maximum of 12ms, and an average of 8ms.
- PC2:** Shows a ping test to 192.168.10.2. The results indicate that 4 packets were sent, 0 were received, and there was a 100% loss. The statistics show a minimum round trip time of 0ms, a maximum of 6ms, and an average of 1ms.
- PC3:** Shows a ping test to 192.168.10.3. The results indicate that 4 packets were sent, 4 were received, and there was a 0% loss. The statistics show a minimum round trip time of 0ms, a maximum of 1ms, and an average of 5ms.

Changing the Native VLAN port to 99 from 10:

The image displays two separate configuration windows for Switch1 and Switch0. Each window shows the configuration commands used to change the native VLAN port from 10 to 99.

Switch1 Configuration:

```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name NATIVE
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#int fa0/24
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)##SPANTREE-2-RECV_FVID_ERR: Received BPDU with inconsistent peer vlan id 1 on FastEthernet0/24 VLAN99.

%SPANTREE-2-BLOCK_FVID_LOCAL: Blocking FastEthernet0/24 on VLAN0099. Inconsistent local vlan.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/24 on VLAN0001. Port consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/24 on VLAN0099. Port consistency restored.
```

Switch0 Configuration:

```
Switch>
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name NATIVE
Switch(config-vlan)#exit
Switch(config)#int fa0/24
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#
```

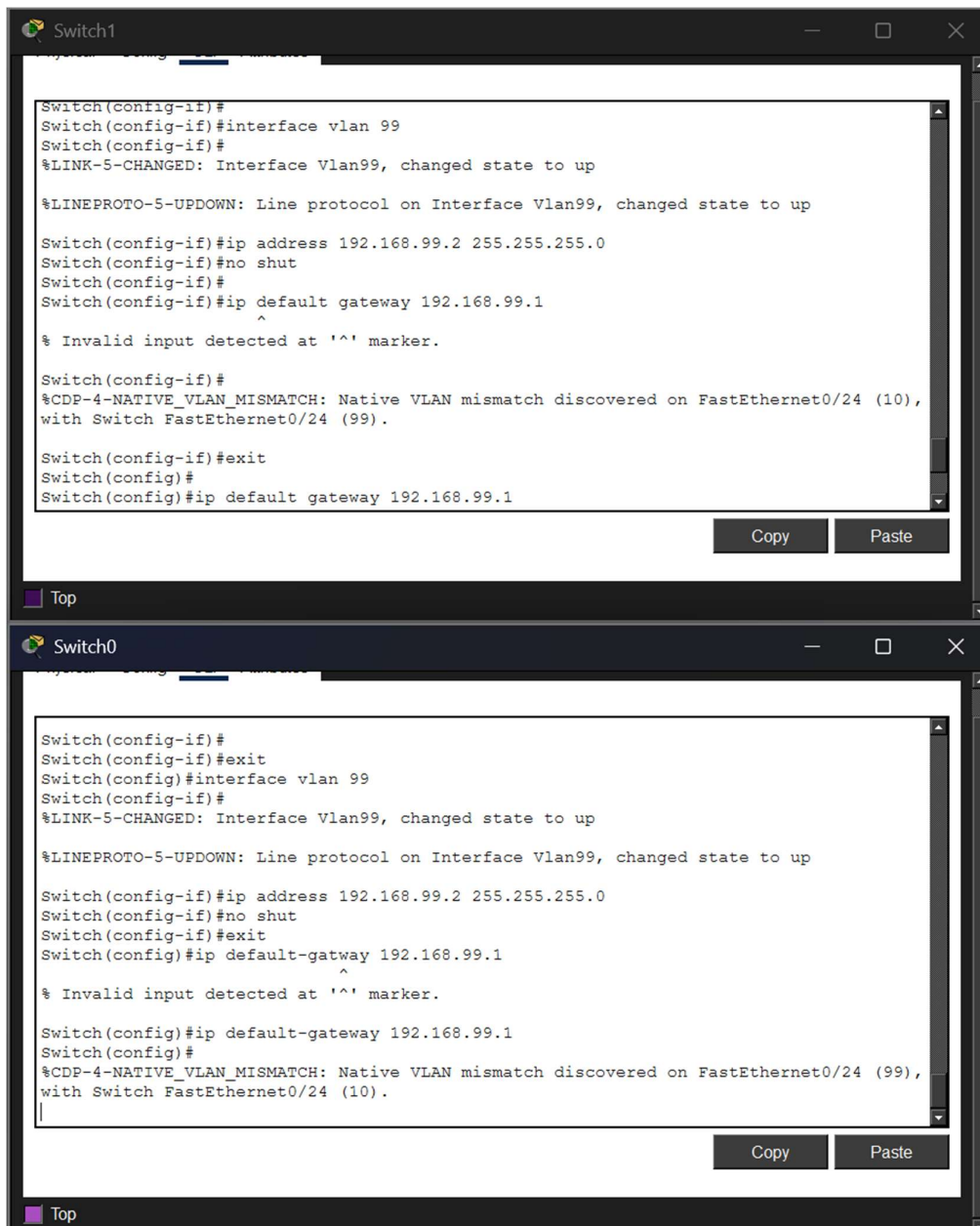

Again, Changing in One switch alone to 10:

```
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport trunk native vlan 10
Switch(config-if)%%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan
id 99 on FastEthernet0/24 VLAN10.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/24 on VLAN0010. Inconsistent
local vlan.

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10),
with Switch FastEthernet0/24 (99).
```

Hence, Troubleshoot:



```

Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0.99
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

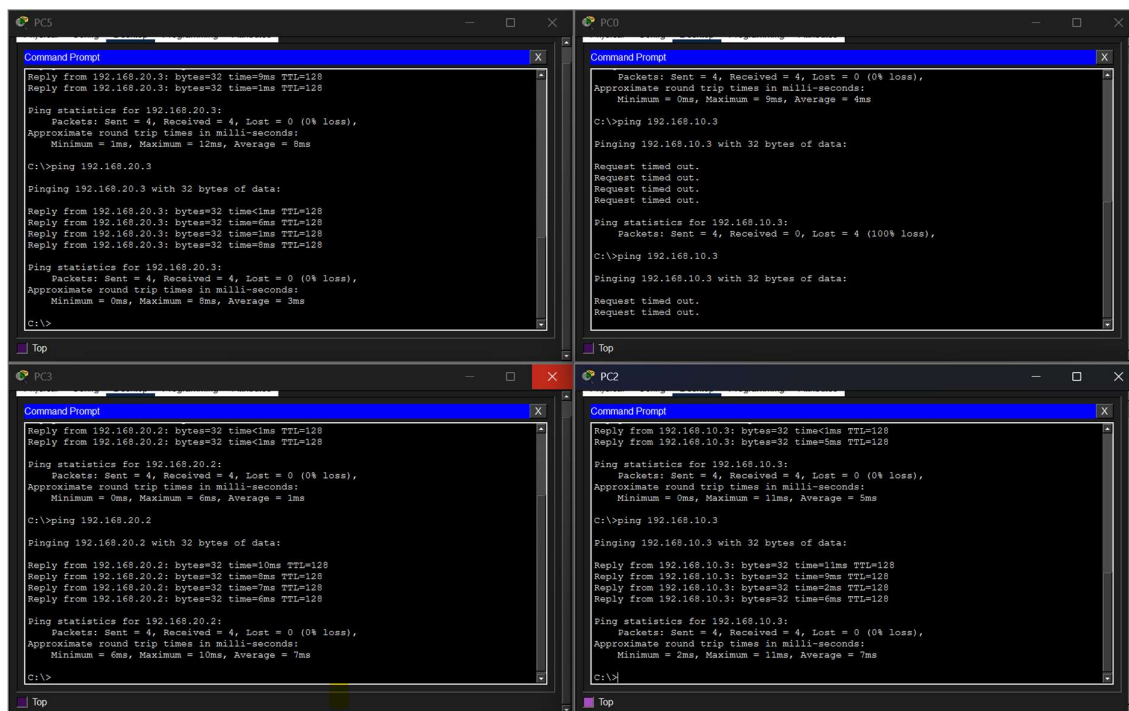
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to
up

Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#

```

Ping Test Once again:



Configure SSH/Telnet:

Switch - 1

```
Switch1
Physical Config CLI Attributes

switch1#
switch1#en
switch1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
switch1(config)#hostname switch1
switch1(config)#ip domain-name lab.local
switch1(config)#crypto key generate rsa
% You already have RSA keys defined named switch1.lab.local .
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: switch1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

switch1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).

*Mar 1 1:25:2.368: %SSH-5-ENABLED: SSH 1.99 has been enabled
switch1(config)#username admin password 1234
switch1(config)#line vty 0 4

% Invalid input detected at '^' marker.

switch1(config)#line vty 0 4
switch1(config-line)#login local
switch1(config-line)#transport input telnet ssh
^
% Invalid input detected at '^' marker.

switch1(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).

switch1(config-line)#
switch1(config-line)#transport input all
switch1(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).
```

Switch - 2:

```
Switch0
Physical Config CLI Attributes

*Mar 1 1:27:43.874: %SSH-5-ENABLED: SSH 1.99 has been enabled
switch0(config)#exit
switch0#
%SYS-5-CONFIG_I: Configured from console by console

switch0#line vty 0 4
^
% Invalid input detected at '^' marker.

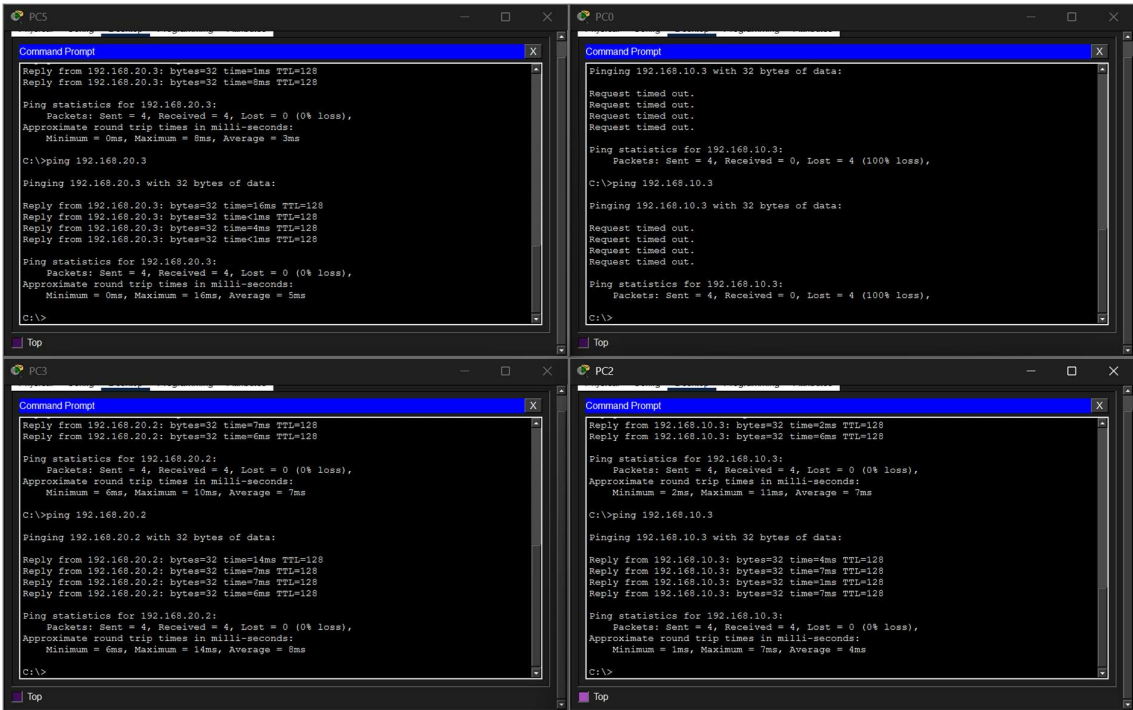
switch0#en
switch0#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
switch0(config)#hostname switch1
switch0(config)#ip domain-name lab.local
switch0(config)#username admin password 1234
switch0(config)#exit
switch0#
%SYS-5-CONFIG_I: Configured from console by console

switch0#en
switch0#con
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).
fig
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
switch0(config)#hostname Switch1
Switch0(config)#ip domain-name lab.local
Switch0(config)#username admin password 1234
Switch0(config)#crypto key generate rsa
% You already have RSA keys defined named switch1.lab.local .
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: Switch1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

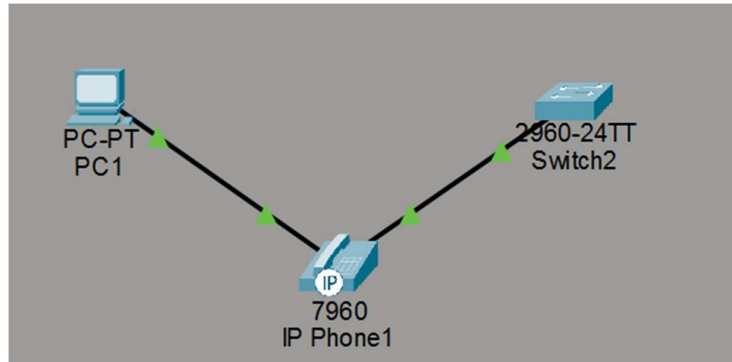
Switch0(config)#line vty 0 4
*Mar 1 1:30:25.365: %SSH-5-ENABLED: SSH 1.99 has been enabled
Switch0(config-line)#transport input all
Switch0(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).
```


Ping Test:



5. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separate VLAN (VLAN 10). Configure the switch port to support both voice and data traffic.

Network Topology:



Switch Config:

```
Switch2
Physical Config CLI Attributes
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name data
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

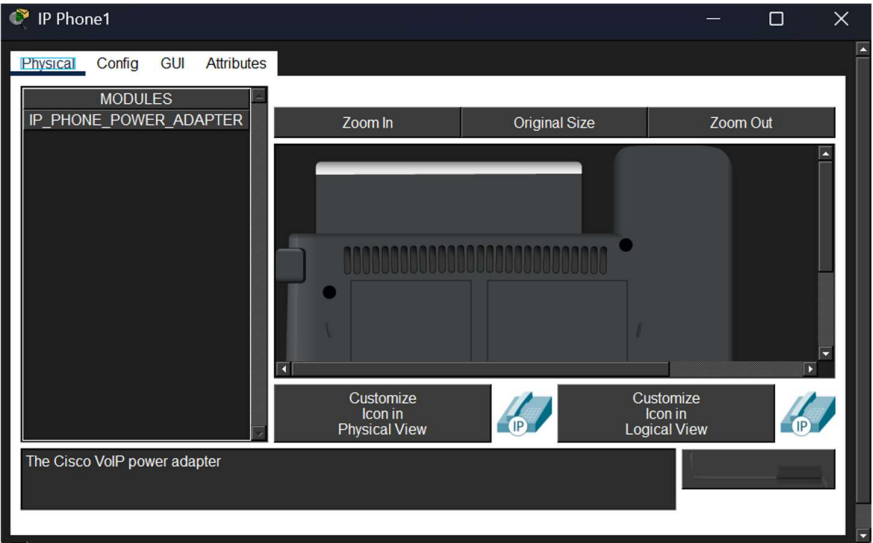
Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name phone
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

Copy Paste

Top

Hence, the adapter to the phone gets created:

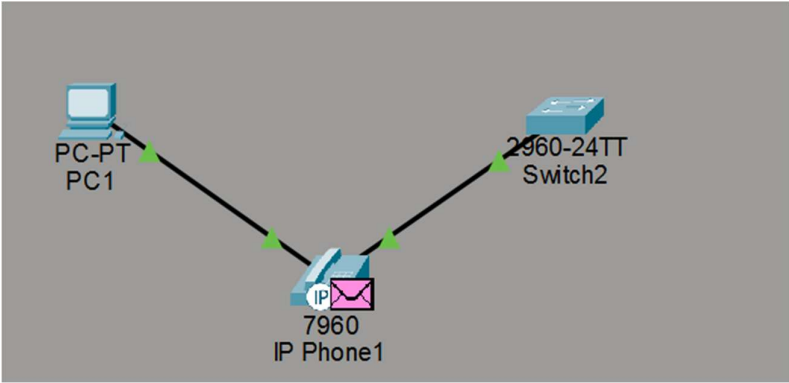


Now, connecting the adapter to the phone:



Hence, the voip phone is powered on and works fine.

Working:



6. You configured VLANs 10 and 20 on your switch and assigned ports to each VLAN. However, devices in VLAN 10 cannot communicate with devices in VLAN 20. Troubleshoot the issue.

In order to enable inter vlan communication, we need a layer 3 device like router and a trunk link configured in it.

Step - 1: Check If VLAN exists with the command: 'sh vlan brief'

Step - 2: Check IP Configuration

- Devices in VLAN 10 → same subnet
- Devices in VLAN 20 → different subnet

Example:

VLAN	IP Range
------	----------

10	192.168.10.0/24
----	-----------------

20	192.168.20.0/24
----	-----------------

Step - 4: Bring Router in:

```
interface fa0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

```
interface fa0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

Step 6: Check Trunk Link

```
show interfaces trunk
```

Ensure VLAN 10 & 20 allowed

Step 7: Verify Default Gateway

Each device must have:

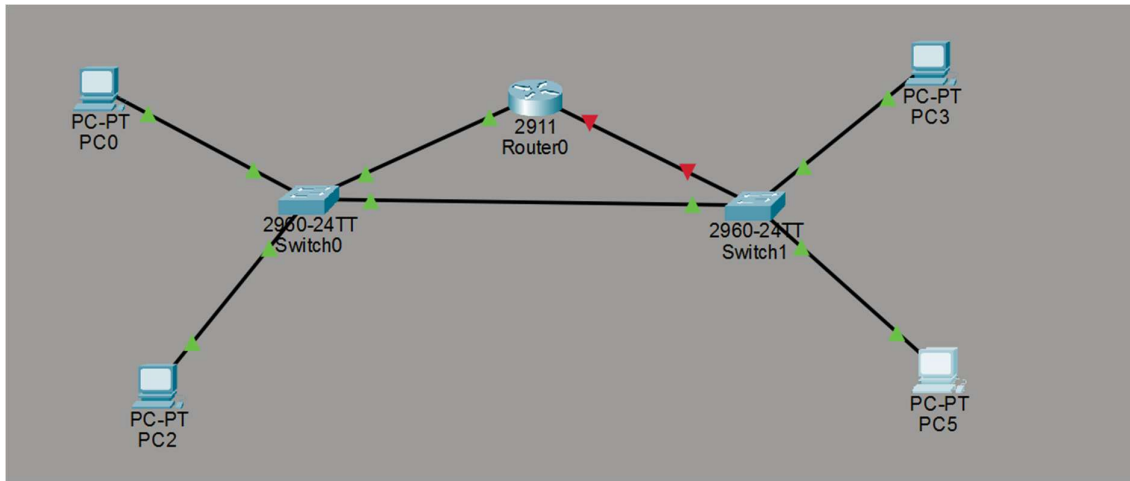
VLAN	Gateway
------	---------

10	192.168.10.1
----	--------------

20	192.168.20.1
----	--------------

7. Try Inter VLAN routing with Router

Network Topology:



Switch - 1 Config:

Set VLAN Identifier Names:

```
Switch5
Physical Config CLI Attributes
Switch>
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
```

Set Port Types for every VLANs:

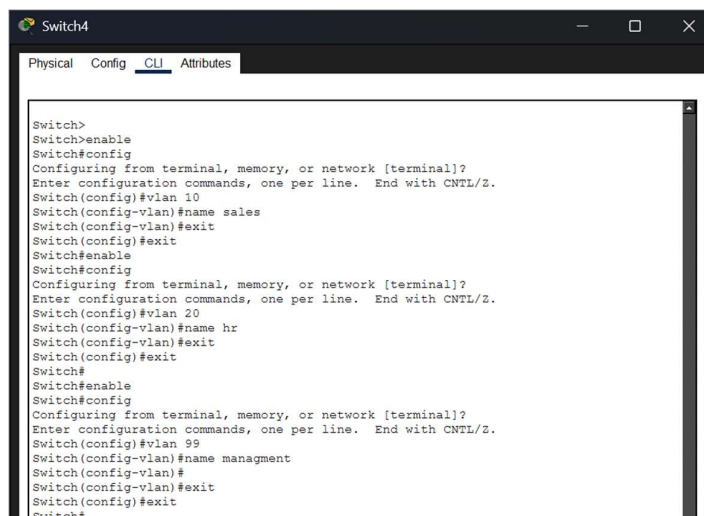
A screenshot of a network switch configuration window titled "Switch6". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI interface shows a series of commands being entered to configure interfaces. The commands are: enable, config, interface fa0/1, switchport mode access, switchport access vlan 10, exit, enable, config, interface fa0/2, switchport mode access, switchport access vlan 20, exit, enable, config, interface fa0/24, switchport mode access, exit, enable, config, interface fa0/2, switchport mode access, exit, enable, config, interface fa0/3, switchport mode trunk, switchport trunk native vlan 99, exit, exit, enable. The window includes a "Copy" button and a "Paste" button at the bottom right, and a "Top" button at the bottom left.

```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
```

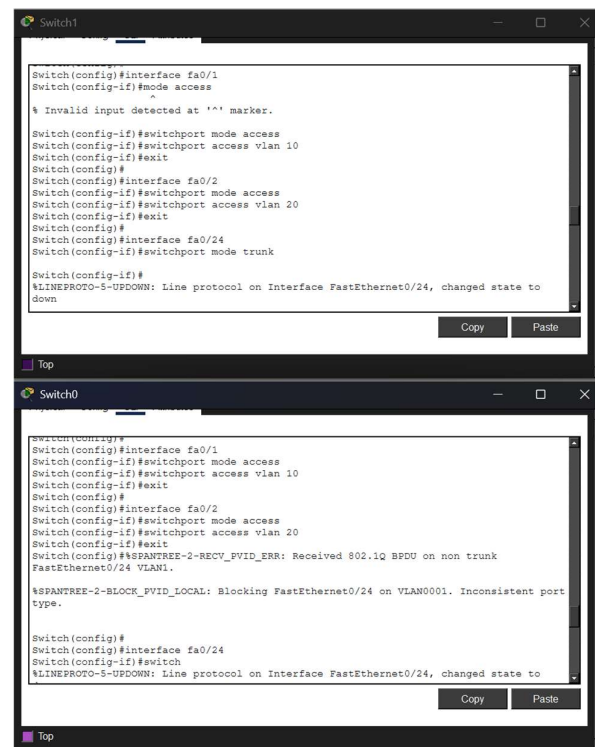
Switch -2 Config:

Create VLAN Identifiers:

A screenshot of a network switch configuration window titled "Switch4". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI interface shows a series of commands being entered to create VLANs. The commands are: enable, config, vlan 10, name sales, exit, enable, config, vlan 20, name hr, exit, enable, config, vlan 99, name management, exit, exit, enable. The window includes a "Copy" button and a "Paste" button at the bottom right, and a "Top" button at the bottom left.

```
Switch>
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#exit
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name management
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
```


Assign Access Ports:



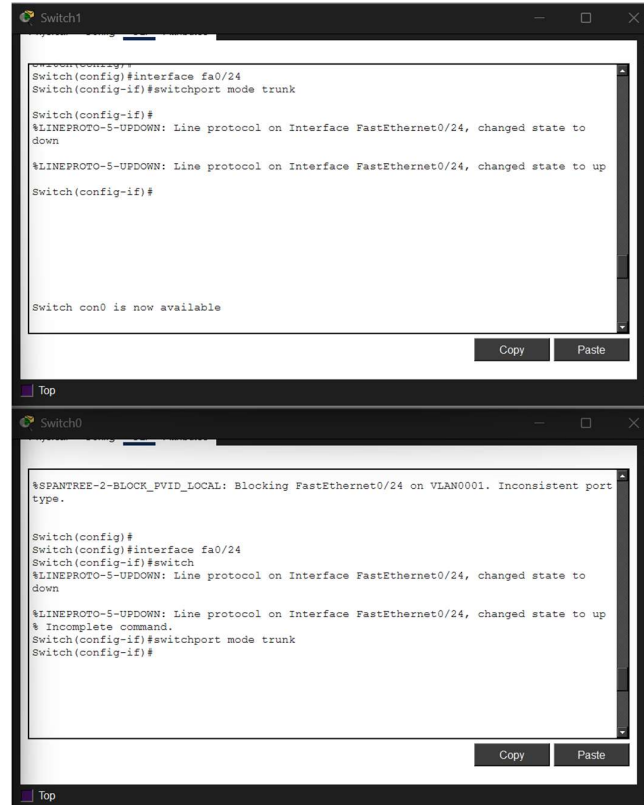
The image shows two terminal windows side-by-side. The top window is titled 'Switch1' and the bottom window is titled 'Switch0'. Both windows show the configuration of access ports. In Switch1, ports fa0/1, fa0/2, and fa0/24 are configured as access ports. In Switch0, ports fa0/1, fa0/2, and fa0/24 are also configured as access ports. The configuration commands are as follows:

```
Switch1
Switch(config)#interface fa0/1
Switch(config-if)#mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch1
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch0
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
```

Assign Trunk Ports:

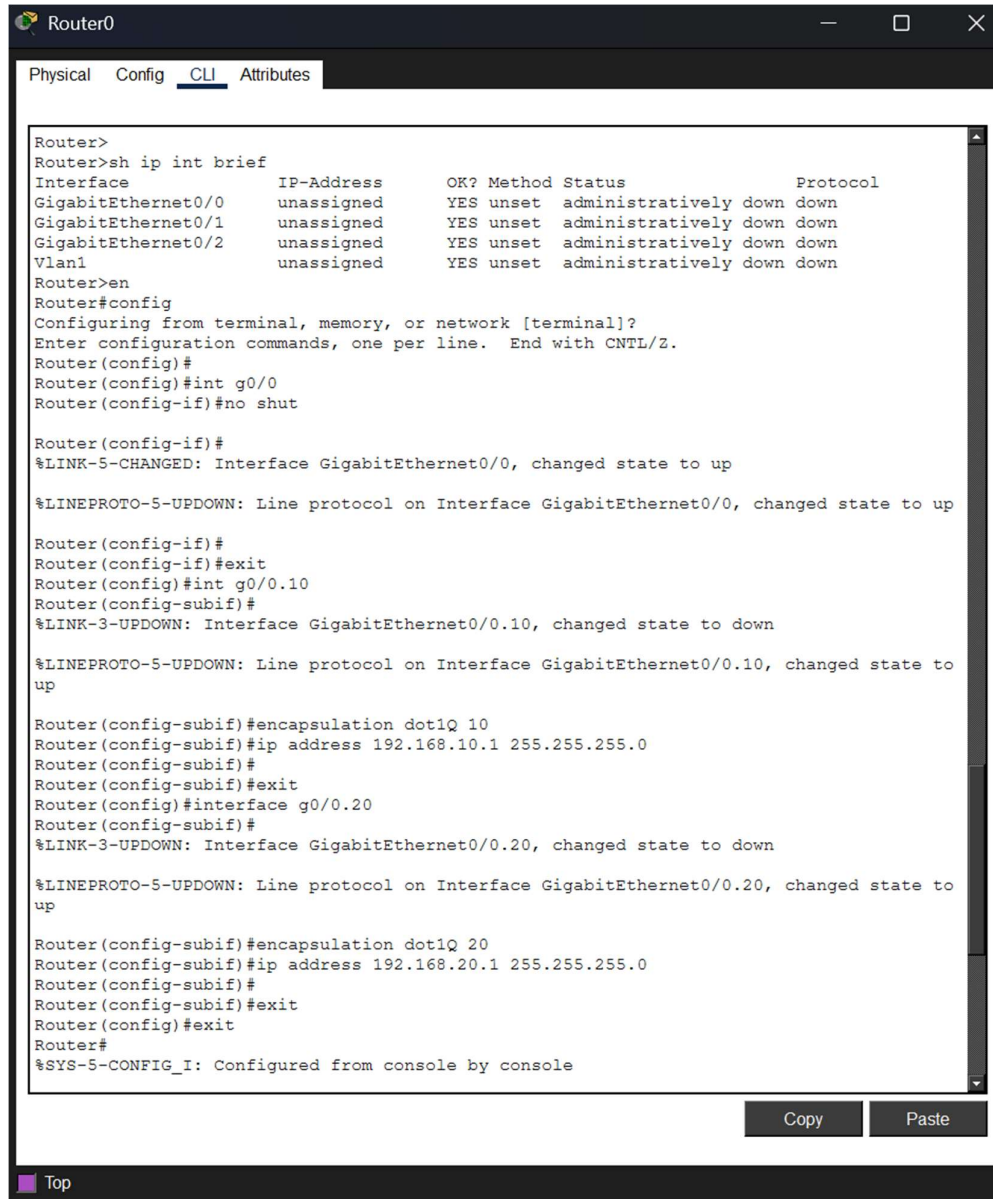


The image shows two terminal windows side-by-side. The top window is titled 'Switch1' and the bottom window is titled 'Switch0'. Both windows show the configuration of trunk ports. In Switch1, port fa0/24 is configured as a trunk port. In Switch0, port fa0/24 is also configured as a trunk port. The configuration commands are as follows:

```
Switch1
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch0
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
```

Router Config:



```
Router>
Router>sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset    administratively down down
GigabitEthernet0/1    unassigned      YES unset    administratively down down
GigabitEthernet0/2    unassigned      YES unset    administratively down down
Vlan1            unassigned      YES unset    administratively down down
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int g0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#int g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

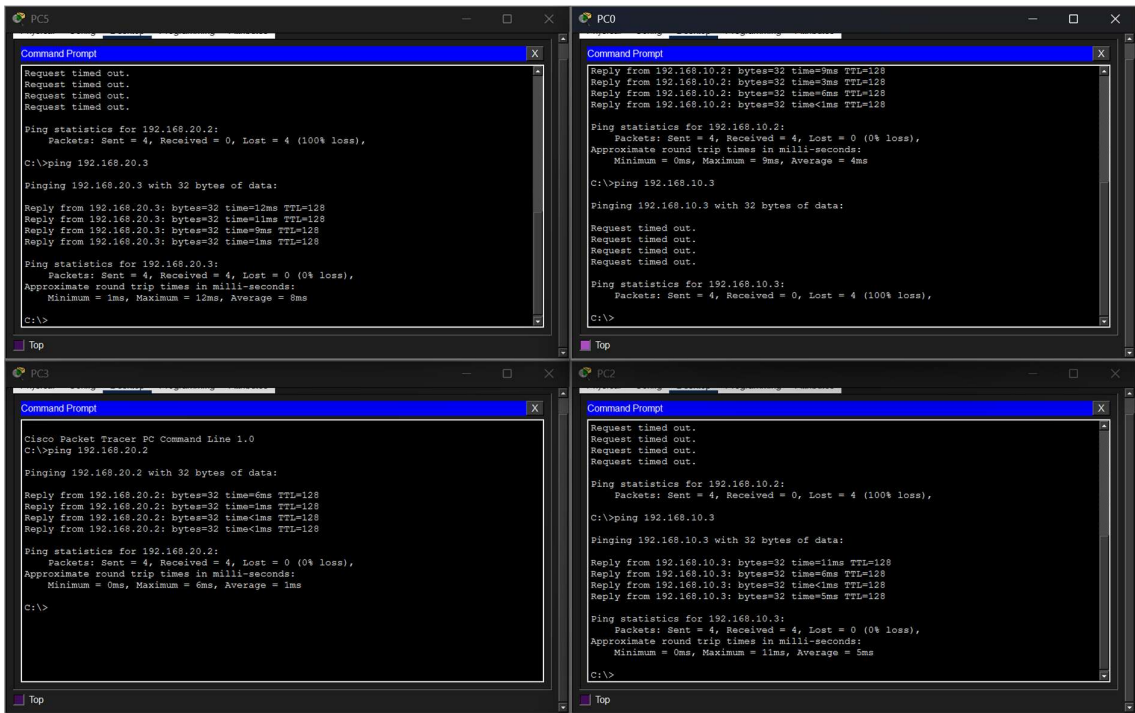
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Configure Router to Switch Port as Trunk Port:

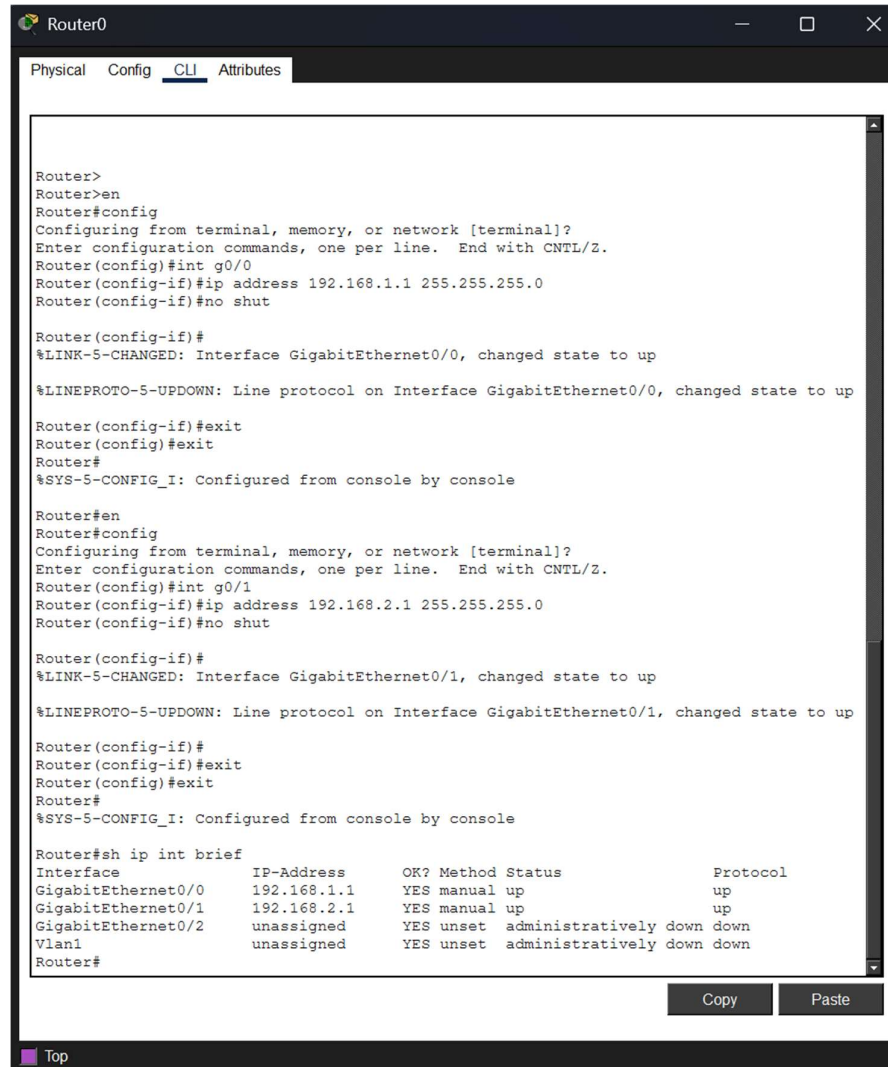
```
Switch>
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping test:



8. Implement ACLs to restrict traffic based on source and destination ports. Test rules by simulating legitimate and unauthorized traffic.

Router Config:



```
Router0
Physical Config CLI Attributes

Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

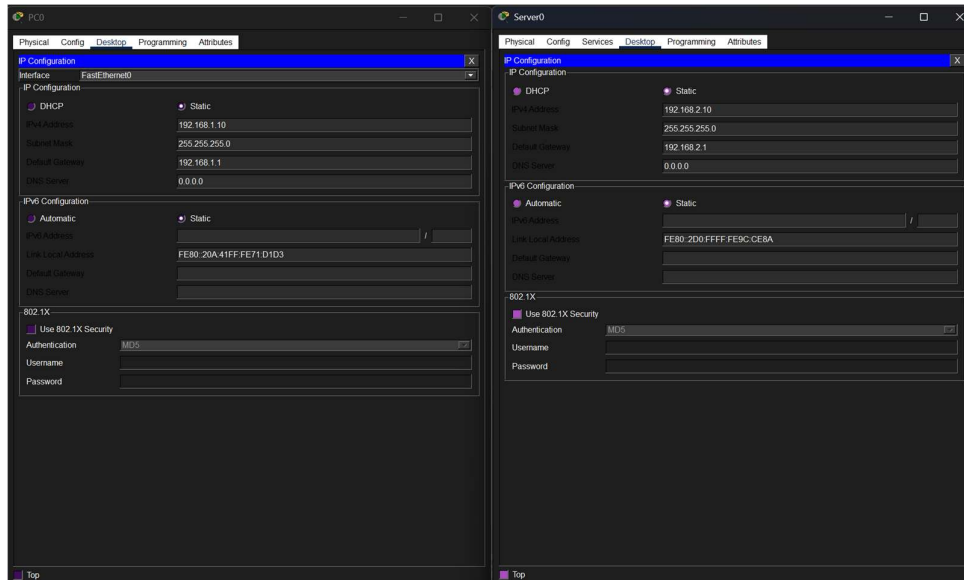
Router(config-if)#exit
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip int brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1     YES manual up          up
GigabitEthernet0/1  192.168.2.1     YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down
Router#
```

Copy Paste

Top

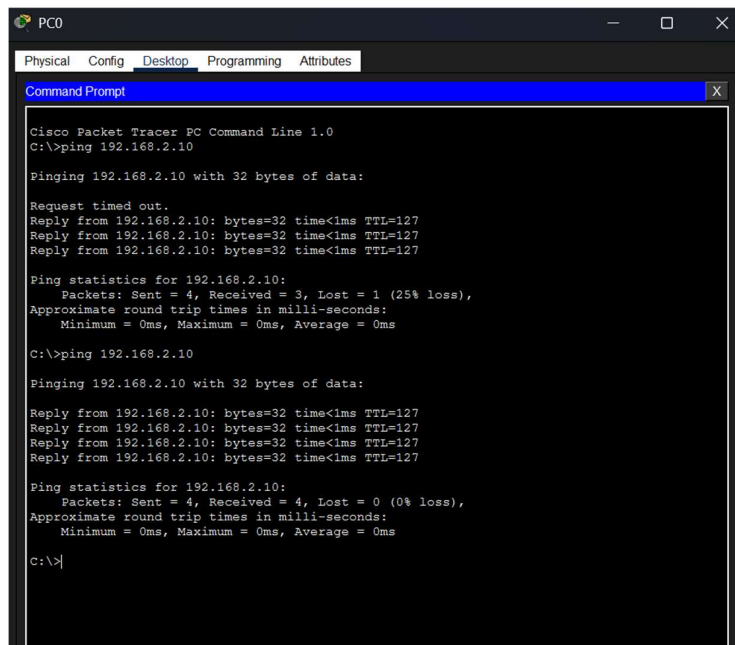
PC Configs:



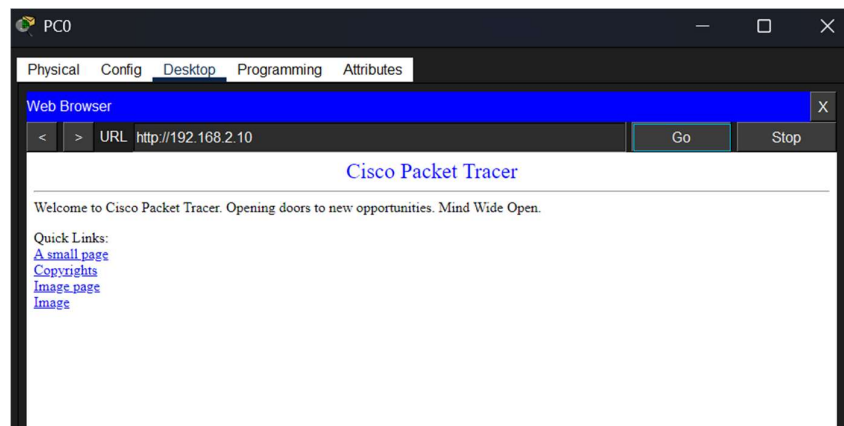
Configure and Enable ACL:

```
Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
Router(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 23
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#int g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

Ping Test from PC to Server:



Checking if the Web Server is Reachable:

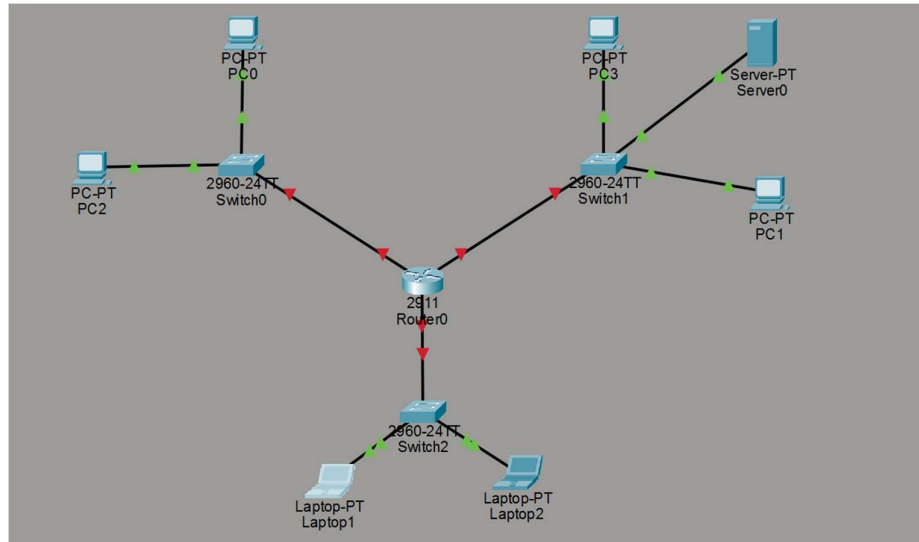


Testing Telnet and Successfully Blocked:

```
C:\>telnet 192.168.2.10
Trying 192.168.2.10 ...
% Connection timed out; remote host not responding
C:\>
```

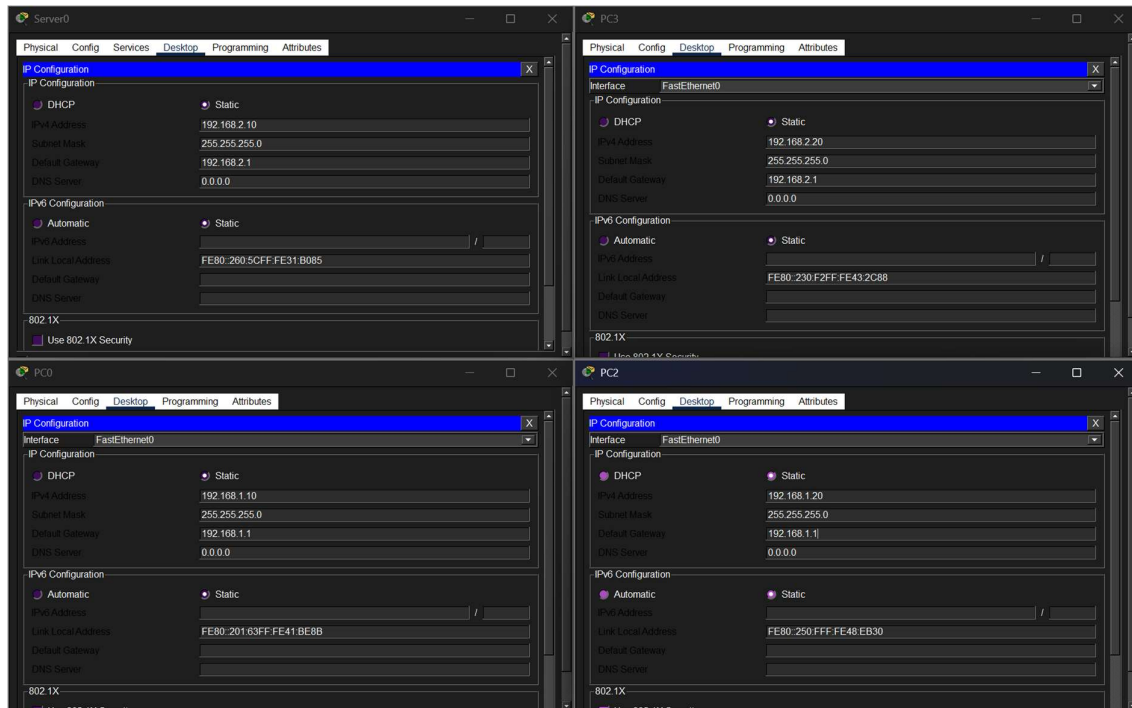
9. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range. Test connectivity to verify the ACL is working as intended.

Network Topology:

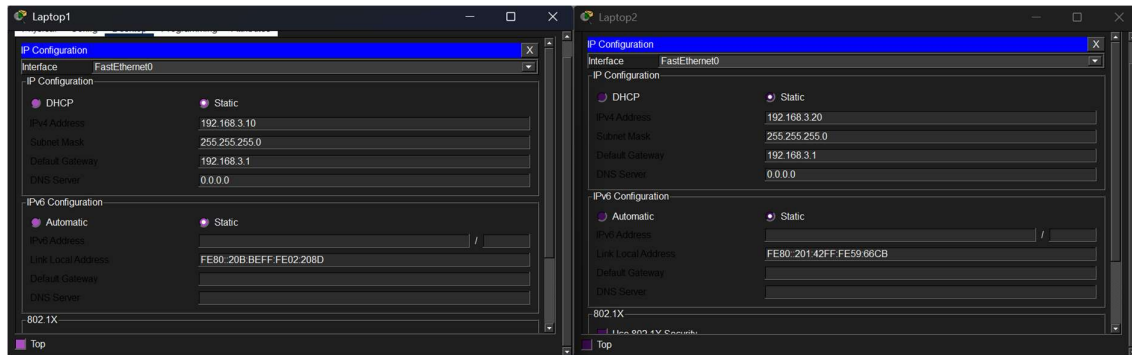


Legal: 192.168.1.0/24, 192.168.20/24 and Outsiders: 192.168.3.0/24

PC Config:



Network Config of Outsider Network PCs:



Router Config:



Routing Table:

```
Router#
Router#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up          up
GigabitEthernet0/1       192.168.2.1     YES manual up          up
GigabitEthernet0/2       192.168.3.1     YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

Creating ACL and Assigning to Interface:

```
Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 permit 192.168.1.0 0.0.0.255
Router(config)#access-list 10 deny any
Router(config)#
Router(config)#ing g0/1
      ^
% Invalid input detected at '^' marker.

Router(config)#int g0/1
Router(config-if)#ip access-group 10 out
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Ping Test from 192.168.1.0/24 network PCs:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms
```

Ping from 192.168.3.0/24 network PCs:

10. Create an extended ACL to block specific applications, such as HTTP or FTP traffic. Test the ACL rules by attempting to access blocked services.

Router Config:

```
R1
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
R1(config)#exit
^
% Invalid input detected at '^' marker.

R1(config)#
*Mar 27 18:20:35.047: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 27 18:20:36.047: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#exit
R1#
R1#
*Mar 27 18:20:37.911: %SYS-5-CONFIG_I: Configured from console by console
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut=ut
^
% Invalid input detected at '^' marker.

R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 27 18:21:09.731: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
*Mar 27 18:21:09.739: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 27 18:21:10.739: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1#

R1#sh ip int brief
Interface                IP-Address      OK? Method Status    Prot
FastEthernet0/0          192.168.1.1     YES manual up        up
FastEthernet0/1          192.168.2.1     YES manual up        up
R1#
```

Config ACL to prevent HTTP and FTP

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 100 deny 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
^
% Invalid input detected at '^' marker.

R1(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 21
R1(config)# 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 21
R1(config)# 100 permit ip any any
R1(config)#
R1(config)#exit
R1#
*Mar 27 18:37:42.703: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Ping Test - 1 Ping within Local LANs:

```
192.168.11.128 - PuTTY
*192.168.1.1 icmp_seq=4 ttl=255 time=12.948 ms (ICMP type:3, code:13, Communication
administratively prohibited)
*192.168.1.1 icmp_seq=5 ttl=255 time=18.814 ms (ICMP type:3, code:13, Communication
administratively prohibited)

PC1> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=80.654 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=46.034 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=17.313 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=9.564 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=16.945 ms

PC1> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=9.901 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=11.190 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=10.245 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=13.413 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=19.252 ms

PC1>

192.168.11.128 - PuTTY
PC2>
PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=45.894 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=24.138 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=15.335 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=21.235 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=9.470 ms

PC2>

192.168.11.128 - PuTTY
PC3>
PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC3> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=63 time=31.378 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=63 time=38.268 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=63 time=34.172 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=63 time=25.584 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=63 time=25.991 ms

PC3>

192.168.11.128 - PuTTY
PC4>
PC4> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=63 time=30.877 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=63 time=30.873 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=63 time=41.365 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=63 time=31.073 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=63 time=26.055 ms

PC4>
```

Telnet Blocked:

```
192.168.11.128 - PuTTY
84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=80.654 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=46.034 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=17.313 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=9.564 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=16.945 ms

PC1> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=9.901 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=11.190 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=10.245 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=13.413 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=19.252 ms

PC1> telnet 192.168.2.10 80
Bad command: "telnet 192.168.2.10 80". Use ? for help.

PC1> telnet 192.168.2.10
Bad command: "telnet 192.168.2.10". Use ? for help.

PC1>

192.168.11.128 - PuTTY
PC2>
PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=45.894 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=24.138 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=15.335 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=21.235 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=9.470 ms

PC2> telnet 192.168.2.10
Bad command: "telnet 192.168.2.10". Use ? for help.

PC2>
```

11. Try Static NAT, Dynamic NAT and PAT to translate Ips

Router Config:

```
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip nat inside
*Mar 27 17:34:46.315: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 27 17:34:47.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#ip nat inside
R1(config-if)#
*Mar 27 17:34:48.839: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
R1#en
*Mar 27 17:34:55.891: %SYS-5-CONFIG_I: Configured from console by console
R1#en c
^
% Invalid input detected at '^' marker.
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 200.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip nat outside
*Mar 27 17:35:29.823: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 27 17:35:30.823: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#ip nat outside
R1(config-if)#
```

PC Configs:

```
192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20002
RHOST:PORT : 127.0.0.1:20003
MTU        : 1500
PC1> █

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.1.20/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
MTU        : 1500
PC2> █
```

Static NAT Config:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip nat inside source static 192.168.1.10 200.1.1.100
R1(config)#
R1(config)#exit
R1#
*Mar 27 17:39:47.263: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 200.1.1.100        192.168.1.10      ---                ---
R1#
```


Configure Dynamic NATs:

Step - 1: Create an ACL

Step - 2: Create a NAT pool consisting some number of Ips

Step - 3: Bind the ACL Pool

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat pool pool1 200.1.1.100 200.1.1.150 netmask 255.255.255.0
R1(config)#
R1(config)#ip nat inside source list 1 pool pool1
R1(config)#
```

PAT:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface f0/1 overload
R1(config)#exit
R1#exit
```

End Router Config Status:

```
R1#sh ip int brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          192.168.1.1    YES manual up            up
FastEthernet0/1          200.1.1.1      YES manual up            up
NVI0                     192.168.1.1    YES unset  up            up
R1#show nat translations
^
% Invalid input detected at '^' marker.

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 200.1.1.100         192.168.1.10     ---               ---
R1#
```

Ping Test:

```
PC1> ping 200.1.1.1

84 bytes from 200.1.1.1 icmp_seq=1 ttl=255 time=20.031 ms
84 bytes from 200.1.1.1 icmp_seq=2 ttl=255 time=26.574 ms
84 bytes from 200.1.1.1 icmp_seq=3 ttl=255 time=21.838 ms
84 bytes from 200.1.1.1 icmp_seq=4 ttl=255 time=10.144 ms
84 bytes from 200.1.1.1 icmp_seq=5 ttl=255 time=25.481 ms
```

12. Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, birectional, reverse, multicast, parallel

Start the Iperf Server:

```
kishore@kishore-virtual-machine:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:71:b7:32 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.31.47/24 brd 192.168.31.255 scope global dynamic noprefixroute ens33
        valid_lft 28433sec preferred_lft 28433sec
    inet6 2409:40f4:1127:484f:f12d:f167:8280:157c/64 scope global temporary dynamic
        valid_lft 16679sec preferred_lft 16679sec
    inet6 2409:40f4:1127:484f:12ac:61da:34da:357/64 scope global dynamic mngtppaddr noprefixroute
        valid_lft 16679sec preferred_lft 16679sec
    inet6 fe80::fe95:eaf3:e714:9839/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kishore@kishore-virtual-machine:~$
kishore@kishore-virtual-machine:~$ iperf3 -s
Server listening on 5201
Accepted connection from 192.168.31.50, port 37920
[ 5] local 192.168.31.47 port 5201 connected to 192.168.31.50 port 37922
[ ID] Interval      Transfer      Bitrate
[ 5] 0.00-1.03 sec  4.62 MBytes  37.8 Mbits/sec
[ 5] 1.03-2.03 sec  7.75 MBytes  64.9 Mbits/sec
[ 5] 2.03-3.00 sec  7.62 MBytes  65.7 Mbits/sec
[ 5] 3.00-4.02 sec  9.50 MBytes  78.6 Mbits/sec
[ 5] 4.02-5.03 sec  5.00 MBytes  41.2 Mbits/sec
[ 5] 5.03-6.02 sec  5.88 MBytes  50.1 Mbits/sec
[ 5] 6.02-7.03 sec  6.00 MBytes  49.8 Mbits/sec
[ 5] 7.03-8.05 sec  7.38 MBytes  60.8 Mbits/sec
[ 5] 8.05-9.11 sec  6.62 MBytes  52.5 Mbits/sec
[ 5] 9.11-10.03 sec  5.50 MBytes  50.0 Mbits/sec
[ 5] 10.03-10.09 sec  384 KBytes  50.4 Mbits/sec
[ ID] Interval      Transfer      Bitrate
[ 5] 0.00-10.09 sec  66.2 MBytes  55.1 Mbits/sec
Server listening on 5201
```

Client Control:

≡ iPerf3

Server address

192.168.31.47

Server port

5201

Streams

1

✓ Upload

Download

Start

52.3 Mbps

Avg: 56.2

Min: 39.8

Max: 81.8

Interval	Transfer	Bi
0.00-1.00s	5.24 MBytes	tr
1.00-2.00s	8.51 MBytes	at
2.00-3.00s	8.25 MBytes	e
3.00-4.00s	10.2 MBytes	41
4.00-5.00s	4.98 MBytes	.8
5.00-6.00s	6.29 MBytes	Mb
6.00-7.00s	6.16 MBytes	ps
7.00-8.00s	7.47 MBytes	68
8.00-9.00s	6.68 MBytes	.2
9.00-10.00s	6.55 MBytes	Mb
		ps
		66
		.0
		Mb
		ps
		81
		.8
		Mb
		ps
		39